

CANONICAL LESSON

Earn the Model's Complexity

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C04
Notebook	V7_0_N04
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Adopt complexity only when it improves future, decision-relevant performance over transparent baselines and remains governable.

Learning outcomes

- Compare transparent baselines with stronger models using time-aware validation, calibration, uncertainty, subgroup checks, and stress tests.
- Select complexity only when it improves decision-relevant performance.

Why this matters

Random splitting, weak baselines, and point estimates can create impressive results that fail under real forecasting conditions.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

1. Time order is part of the evidence

- Random splitting leaks future structure into training.
- Evaluation must simulate the real forecasting origin and horizon.

2. Use several transparent baselines

- Last observation, seasonal history, and trend encode different assumptions.
- A sophisticated model must outperform relevant baselines, not a deliberately weak comparator.

3. A stronger combined model

- This illustrative regression combines trend and seasonal terms.
- Its value is empirical: it must earn adoption on held-out future data.

4. Prediction intervals and empirical coverage

- Point forecasts conceal uncertainty.
- Interval quality requires both coverage and useful width, evaluated on future observations.

5. Stress testing

- Test plausible disruption, missing seasonality, changed reporting delay, and extreme values.
- A model can be accurate on average yet fragile under policy-relevant conditions.

6. Complexity decision card

- Performance is necessary but not sufficient.
- Consider maintainability, latency, explainability, calibration, subgroup harm, and retirement.

Worked cross-sector example

A seasonal regression is compared with last-value, seasonal-history, and trend baselines using time-ordered holdouts and interval coverage.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Add rolling-origin evaluation.
- 2.
- Compare MAE and RMSE consequences.
- 3.
- Recalibrate the interval.
- 4.
- Define a retirement trigger.

Answer key

- 1.
- Refit/forecast at successive historical origins and aggregate horizon-specific metrics.

- 2.
- RMSE penalizes large errors more; choose according to decision loss rather than habit.
- 3.
- Estimate residual quantiles on a calibration window and verify held-out coverage and width.
- 4.
- Examples: persistent baseline underperformance, failed coverage/calibration, unacceptable subgroup harm, unmaintainable dependencies, or changed purpose.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.